Lottery Ticket Analysis

Math 216B

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Code: `https://github.com/qndo1/math216B/tree/main/Final_proj`

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Abstract

This project explores two questions related to the Lottery Ticket Hypothesis (LTH): whether learned pruning masks transfer across datasets and whether the emergence of lottery tickets is facilitated when the underlying function depends on a small subset of parameters. For the transferability study, pruning masks learned from models trained on CIFAR-10 were evaluated on CIFAR-100 and compared against randomly generated masks. The results suggest that lottery ticket masks do not consistently produce networks that outperform dense baselines on different datasets. However, there is evidence that masks learned through training generally perform better than random masks when transferred, indicating that the pruning structure captures some reusable features of the network.

For the functional structure study, synthetic datasets were generated to approximate single-index and multi-index models, and performance was evaluated using mean squared error (MSE). Experiments indicate that lottery ticket subnetworks are more easily discovered and perform more reliably in settings where the target function depends on a single linear projection of the input. In contrast, multi-index functions show greater degradation as networks become increasingly sparse. These results suggest that the success of lottery tickets may depend in part on the intrinsic dimensionality of the function being learned, particularly in resource-limited training regimes.

1. Introduction

Modern neural networks are often heavily overparameterized, yet empirical evidence shows that many parameters can be removed without significantly affecting performance. The Lottery Ticket Hypothesis (LTH), introduced by Frankle and Carbin [1], proposes that within randomly initialized dense neural networks there exist sparse subnetworks, referred to as *winning tickets*, that can be trained in isolation to achieve comparable performance to the original network. These subnetworks are identified by pruning weights after training and resetting the remaining weights to their original initialization. Subsequent work has further explored the properties of these subnetworks and the conditions under which they emerge.

Despite strong empirical support for the existence of lottery tickets, several questions remain regarding their generality and the mechanisms behind their success. In particular, it is unclear whether the pruning masks that define these subnetworks capture transferable structure across tasks, or whether they are specific to the dataset on which they were discovered. Additionally, it is not well understood how the underlying structure of the target function affects the ease with which lottery ticket paths emerge.

In this project, we investigate two related questions. First, we study whether pruning masks learned on one dataset transfer effectively to another. Specifically, we train networks on CIFAR-10, extract pruning masks, and evaluate their performance when applied to CIFAR-100. These results are compared against randomly generated masks to assess whether learned masks capture useful transferable structure. Second, we examine whether the emergence of lottery tickets is influenced by the intrinsic dimensionality of the target function. Using synthetic datasets, we compare performance on single-index functions of the form $f(x) = g(\langle w, x\rangle)$ and multi-index functions that depend on multiple projections of the input. Performance is evaluated using mean squared error (MSE) while progressively increasing network sparsity.

Our results suggest that while learned pruning masks do not necessarily produce optimal networks for different datasets, they tend to outperform random masks when transferred. Furthermore, lottery ticket subnetworks appear easier to discover and maintain performance for single-index functions, while multi-index functions degrade more rapidly as sparsity increases.

2. Methods

2.1. Model Architecture

For the image classification experiments, we use a convolutional neural network consisting of three convolutional layers followed by two fully connected layers. Each convolutional layer uses 3×3 kernels with padding and is followed by a ReLU activation and 2×2 max pooling. The convolutional layers produce feature maps of width $w = 128$, after which the representation is flattened and passed through a fully connected layer with 256 hidden units and a final classification layer. The network is trained using the Adam optimizer with a learning rate of 10^{-3} and a batch size of 512.

2.2. Datasets

To study the transferability of lottery ticket subnetworks, we use the CIFAR-10 and CIFAR-100 datasets. CIFAR-10 consists of 60,000 32×32 RGB images across 10 classes, while CIFAR-100 contains the same number of images distributed across 100 classes. All images are converted to tensors without additional augmentation. Training and evaluation are performed using standard training and test splits for each dataset.

2.3. Lottery Ticket Identification

To identify lottery ticket subnetworks, we follow an iterative magnitude pruning procedure. A dense network is first initialized and trained on CIFAR-10 for two epochs. After training, a global pruning step removes the lowest-magnitude weights across all layers. Specifically, at each pruning round the bottom 20% of remaining weights (by absolute value) are removed. The remaining weights define a binary pruning mask.

Following the Lottery Ticket Hypothesis procedure, the surviving weights are reset to their original initialization values while preserving the pruning mask. The masked network is then retrained and pruned again. This process is repeated for ten rounds, progressively increasing the sparsity of the model while tracking classification accuracy.

2.4. Transfer Experiment

To evaluate whether lottery ticket masks transfer across datasets, the final pruning mask obtained from training on CIFAR-10 is applied to a newly initialized network trained on CIFAR-100. Performance is compared against two baselines:

- A dense CIFAR-100 model with no pruning.
- A randomly pruned model with the same sparsity level as the transferred mask.

Each model is trained for ten epochs using the Adam optimizer, and classification accuracy on the CIFAR-100 test set is recorded. Experiments are repeated multiple times to estimate average behavior across runs.

2.5. Synthetic Function Experiments

To investigate how the intrinsic dimensionality of the target function affects the emergence of lottery tickets, we conduct experiments using synthetic datasets. Inputs $x \in \mathbb{R}^d$ are sampled from a standard Gaussian distribution with $d = 100$.
Two function classes are considered:

- **Single-index model:** $f(x) = \sin(\langle w, x \rangle)$, where w is a random vector.
- **Multi-index model:** $f(x) = \sum_{i=1}^{k} \sin(\langle w_i, x \rangle)$ with $k = 10$ independent projections.

For these experiments we train a fully connected neural network with two hidden layers of width 512 and ReLU activations. The network outputs a scalar prediction and is trained using mean squared error (MSE) loss with the Adam optimizer.

2.6. Pruning Procedure for Synthetic Experiments

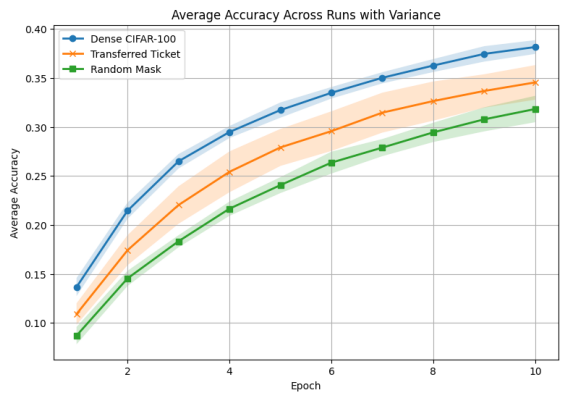
After training a dense model, we apply global magnitude pruning at various sparsity levels ranging from 20% to 95%. Pruned weights are removed by applying binary masks and the remaining weights are reset to their initial values. Each masked network is then retrained from initialization using the same training procedure as the dense model.

Model performance is evaluated by computing the mean squared error on the training dataset. We compare the behavior of pruned subnetworks across sparsity levels for both single-index and multi-index functions to analyze how intrinsic function complexity affects the emergence of lottery tickets.

3. Results

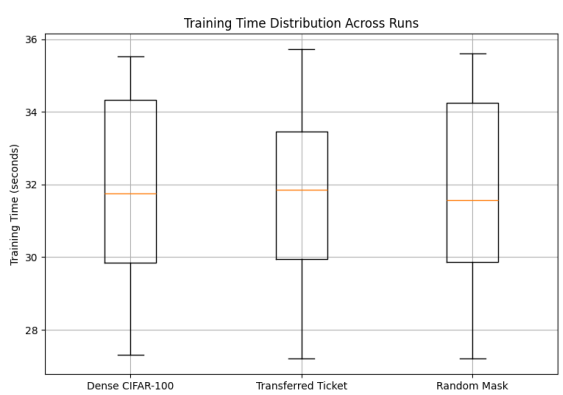
3.1. Transferability of Lottery Ticket Masks

We first evaluate whether pruning masks learned on CIFAR-10 transfer effectively to a different dataset. Specifically, masks identified through iterative magnitude pruning on CIFAR-10 were applied to networks trained on CIFAR-100. Performance was compared against two baselines: a dense network trained on CIFAR-100 and a randomly pruned network with the same sparsity level as the transferred mask.



Across repeated experiments, dense networks consistently achieved the highest classification accuracy. Networks using transferred masks performed worse than the dense baseline but generally outperformed networks with randomly generated masks. This suggests that the pruning structure learned on CIFAR-10 captures some useful parameter configuration that remains beneficial when applied to a related dataset, even though it does not fully match the performance of an unpruned model.

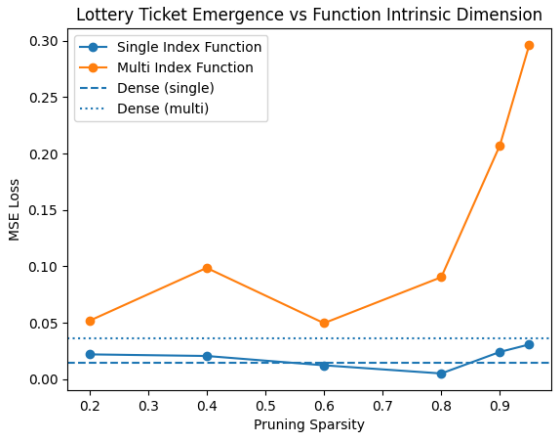
We also observed that masked networks exhibited noticeably higher variance in test accuracy across runs compared to the dense baseline. Both transferred masks and random masks produced more variable results, indicating that sparsity introduces greater sensitivity to initialization and training dynamics. The dense networks, by contrast, showed more stable performance across trials.



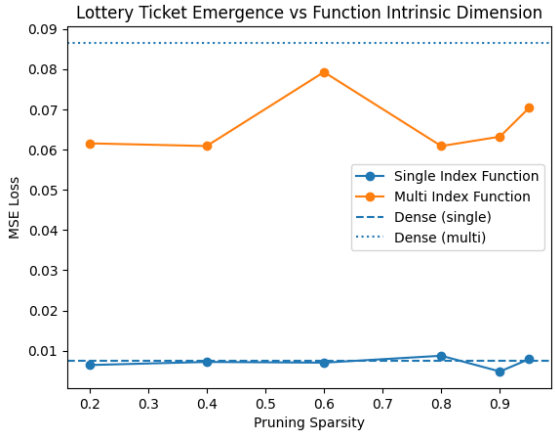
In addition to accuracy, we recorded the time required to complete the 10 training epochs for each model configuration. All experiments were conducted on an NVIDIA 2080 Ti GPU using PyTorch's CUDA support. While the code provided in the accompanying GitHub repository can also run on CPU, runtimes in that setting are significantly longer and may exceed one hour. As shown in the figure above, the overall training times were similar for the dense network, the transferred mask network, and the randomly masked network. This behavior is likely due to the implementation of pruning used in our experiments: masks are applied to a fully dense network rather than using a specialized sparse computation framework. As a result, the underlying matrix operations remain dense, preventing the training procedure from realizing potential computational speedups that could arise from exploiting the sparse structure of the pruned networks.

3.2. Effect of Intrinsic Function Dimension

We next examined how the intrinsic dimensionality of the target function influences the emergence and effectiveness of lottery ticket subnetworks. Synthetic datasets were generated using both single-index functions and multi-index functions, and models were evaluated using mean squared error (MSE) under varying sparsity levels.

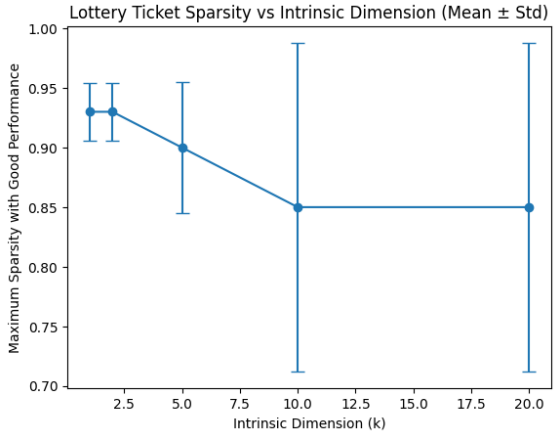


When training was performed with relatively small numbers of epochs, a clear divergence emerged between the single-index and multi-index settings at high sparsity levels. Networks trained on single-index functions maintained relatively low error even as a large fraction of weights were pruned. In contrast, models trained on multi-index functions experienced significant degradation in performance as sparsity increased. This suggests that when the underlying function depends on a small number of directions in the input space, sparse subnetworks are more likely to capture the relevant structure.



However, when the number of training epochs was increased, the performance gap between the two settings diminished. With sufficient training time, both single-index and multi-index models were able to converge

to similar MSE values across many sparsity levels. This indicates that the advantages of lottery ticket subnetworks for low intrinsic dimensional functions are most pronounced in limited training regimes, where computational resources or training time are constrained.



To further examine this relationship, we evaluated networks across a range of intrinsic dimensions using k-index synthetic functions. For each dimension, we measured the maximum sparsity level that achieved an error within a fixed threshold of the dense model. Results showed that higher intrinsic dimensions exhibited substantially greater variability in the sparsity levels that satisfied this threshold. In general, functions with larger intrinsic dimension required denser networks to maintain comparable performance, reinforcing the observation that the effectiveness of lottery tickets depends in part on the complexity of the underlying function.

4. Conclusion

In this report, we investigated two aspects of the Lottery Ticket Hypothesis: the transferability of pruning masks across datasets and the relationship between intrinsic function dimensionality and the emergence of sparse subnetworks. Our experiments show that pruning masks learned on CIFAR-10 can transfer to CIFAR-100, generally outperforming random masks, but still falling short of the accuracy achieved by dense networks. We also observed that masked networks exhibit higher variance in performance, reflecting increased sensitivity to initialization and training dynamics.

For synthetic datasets, we found that functions with low intrinsic dimensionality, such as single-index models, benefit more from sparse lottery ticket subnetworks, particularly in low-resource or limited-epoch training regimes. Multi-index functions, by contrast, are more sensitive to pruning, although given sufficient training epochs, the performance of sparse networks converges to that of dense models. Across increasing intrinsic dimensions, higher-dimensional functions tolerate less sparsity for a fixed error threshold, indicating that the effectiveness of lottery tickets is closely tied to the complexity of the underlying function.

Overall, these findings highlight that while lottery ticket subnetworks capture reusable structure in both images and synthetic functions, their practical benefits are most pronounced in settings with limited training resources or when the target function depends on a small subset of parameters. Future work could explore specialized sparse training methods to reduce computation and further enhance the advantages of pruned networks.

References

[1] Jonathan Frankle and Michael Carbin. The lottery ticket hypothesis: Training pruned neural networks. *CoRR*, abs/1803.03635, 2018.